How Do World Action Models Use Their Futures? Evidence from Mechanistic Interventions

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Abstract

Do world action models actually use the futures they generate? Jointly producing future observations and robot actions does not answer this question: the action may bypass the predicted future. We test future use with reachable-donor transplantation. Two native policy runs begin from the same restored simulator state and produce different futures, actions, and endpoints. We insert one run's future into the other while holding the current observation, instruction, sampling noise, and denoising schedule fixed. If future content matters, the recomputed action should move specifically toward the action associated with the donor future. In Cosmos 3, a frozen study across 22 states and six RoboLab tasks finds predicted-future action steering of 0.992 and executed-future endpoint steering of 1.003 on a normalized recipient-to-donor scale. Replacing donor-run future-token keys and values with self-future counterparts reduces predicted-action steering by 0.877, with positive action and endpoint reductions in all 22 states. Isolated robot- and object-pixel interventions, however, each have effects below 0.011. Cosmos Policy shows a related early-state effect across all ten LIBERO-Long tasks, carried mainly by visual content. These results show that both direct policies use coherent futures to condition action, but do not establish candidate search or planning over predicted outcomes.

1 Introduction

Many world action models (WAMs) generate future observations alongside robot actions [8, 14]. It is tempting to read this architecture literally: the model predicts what may happen, then acts on that prediction. Joint generation does not establish such a computation. The future could be irrelevant to the action, serve as a generic workspace, encode the robot motion already paired with the action, or represent task consequences used to choose among actions.

These alternatives imply different functional roles. A policy that conditions action inference on predicted future observations resembles an inverse-dynamics interface [1, 7]. A system that generates alternatives and compares their predicted outcomes with a cost or learned heuristic more closely resembles explicit visual planning [2, 3]. We study Cosmos Policy's direct-policy mode and the diffusion-generator component of Cosmos3-Nano-Policy-DROID. The inference procedures evaluated here do not generate and score multiple candidate actions before acting; Cosmos Policy's separate best-of-N planning mode is outside our study.

RIFT provides the closest causal evidence that future representations matter [22]. Masking the future read or editing values stored at future positions changes closed-loop trajectories and task success across several WAMs. This establishes dependence on a structured future interface, but a destructive intervention does not say what a particular coherent alternative future tells the policy to do. A damaged representation can move an action in any direction.

Our test is directional. From the same restored state, two native policy continuations define a recipient and a donor. We insert the donor future into the recipient computation while retaining the recipient's current inputs and random draws. The model never receives the donor action; we use

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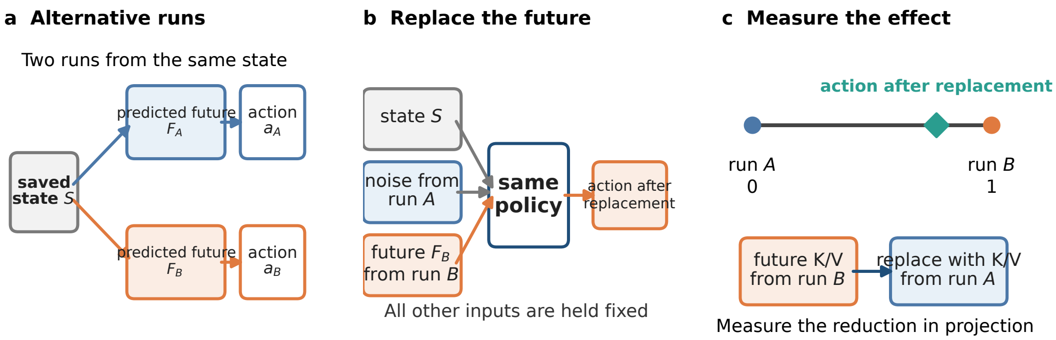

Figure 1: Reachable-donor future transplantation. Two unmodified continuations start from the same simulator state. The recipient is recomputed with the donor future while its other inputs and random draws remain fixed. Signed projections measure movement from recipient behavior (0) toward donor behavior (1); K/V replacement then tests whether future-token activations carry that shift.

it only to define the direction in which we score the recomputed action and its executed simulator endpoint (Figure 1). A generic perturbation need not move either quantity in that direction. A future whose content controls behavior should.

We ask three questions: Do coherent donor futures steer behavior? Which parts of the future recreate that effect? Which future-to-action interfaces carry it? Cosmos 3 provides the strongest within-model chain: steering generalizes across 22 sampled states, isolated robot or object pixels do not reproduce it, and future-token K/V replacement suppresses action and endpoint steering in every state. Cosmos Policy supplies an earlier-model comparison whose effects are concentrated at early states and are most consistent with camera-visible robot motion. The evidence supports future-dependent action generation, but not candidate search, outcome comparison, or task-level planning.

2 Related Work

Prediction and control. Some visual robot policies generate future frames and infer actions from them; others condition action inference on predicted video representations [1, 7]. Joint video–action models instead learn future observations and controls in one model [6, 8, 10, 14, 20]. Their training objective does not determine whether a particular generated future is used at inference. UVA, for example, can omit video generation at test time [10]. Explicit visual planners answer a different question by sampling alternatives, predicting their outcomes, and comparing them with a goal-dependent score [2, 3].

Recent WAM studies separate predictive training from the use of predictions during action generation. Fast-WAM removes explicit test-time rollout while retaining competitive performance in its evaluations [21]; Faster-WAM retains a lower-cost future-conditioning route and reports improved robustness [24]. AGRA finds that visually plausible foresight need not support reliable action inference in its studied architecture [16]. RIFT directly establishes dependence on future-position values and their organization [22]. Our contribution is complementary: we exchange two coherent, policy-generated alternatives and ask whether behavior follows the selected donor.

Mechanistic interventions. Activation replacement tests whether an internal representation affects an output rather than merely being decodable from it [12, 18]. Path patching extends this logic to specified computational routes [5]. Conclusions can depend on the counterfactual pair and metric [23], while multi-component effects need not add when patched and unpatched routes interact [17]. We therefore treat K/V replacement as evidence for an active pathway, not a complete circuit decomposition.

3 Methods

3.1 Future transplantation and directional projection

Let saved simulator state S admit two unmodified branches A and B, with future representations F_A, F_B, action chunks $\mathbf{a}_A, \mathbf{a}_B$, and endpoints $\mathbf{e}_A, \mathbf{e}_B$ obtained by executing one action chunk. We call A the recipient and B the donor. To recompute A, we restore the same state and observation, retain its instruction, action noise, future-noise path, solver, and denoising schedule, and replace only F_A with F_B. We also reverse the branch roles.

For an action or endpoint $\mathbf{x} \in \{\mathbf{a}, \mathbf{e}\}$, normalized donor projection is

$$\pi_x(\hat{\mathbf{x}}; A, B) = \frac{(\hat{\mathbf{x}} - \mathbf{x}_A)^\top (\mathbf{x}_B - \mathbf{x}_A)}{\|\mathbf{x}_B - \mathbf{x}_A\|_2^2}. \tag{1}$$

The recipient is 0 and the donor is 1; negative values move away from the donor and values above 1 overshoot it. The primary steering contrast subtracts the corresponding self-future recomputation:

$$\Delta_x = \pi_x(\hat{\mathbf{x}}_{\mathrm{donor}}; A, B) - \pi_x(\hat{\mathbf{x}}_{\mathrm{self}}; A, B). \tag{2}$$

Pairs are selected using native action and endpoint separation, never intervention outcomes. Every endpoint arm is executed after restoring the exact simulator state.

3.2 Models, populations, and controls

Cosmos Policy. We evaluate the public `Cosmos-Policy-LIBERO-Predict2-2B` checkpoint in direct-policy mode on LIBERO-Long, also called LIBERO-10 [8, 11]. The frozen grid contains 30 states: ten tasks crossed with the first query, a state after three open-loop chunks, and a state after six chunks. Eight native runs determine one reciprocal donor pair per state; three future-noise draws are repeated within direction. Endpoints concatenate goal-relevant simulator coordinates with the official nine-dimensional proprioceptive endpoint.

Cosmos 3. We evaluate the 16B `Cosmos3-Nano-Policy-DROID` checkpoint on six eligible RoboLab tasks [14, 15, 19]. Its diffusion generator has 36 two-way full-attention layers, four denoising calls, 33 raw video frames mapped to nine VAE latent frames, and 32 absolute joint-position actions. The transplant targets unconditioned future-video coordinates at the final guided-sampler boundary. A frozen screen selected 24 state clusters, capped at four per task, using only native success, action and endpoint separation, and replay validity. Twenty-two completed. Two source branches ended before producing full 33-frame videos and were left missing rather than replaced. Reciprocal directions and predicted versus executed future sources are repeated measurements within state.

Controls include exact self-future recomputation, Gaussian targets matched to the donor in norm and recipient distance, a prespecified natural-future control, within-task shuffled futures, and bitwise no-op checks. For Cosmos 3, identity layout capture, sampler wrapping, cache record/replay, and self patches reproduce actions bitwise; sentinels confirm that the transplant does not directly write to action coordinates. Executed-future conditions encode video from reachable donor actions and replay the same environment prefix before intervention.

3.3 Robot and object interventions

We isolate content in complementary ways. In Cosmos Policy, first-query interventions replace the wrist-camera future, primary-camera future, or future proprioception separately. A second study crosses binary robot and object states in re-rendered videos at 20 held-out states. A decoder checks whether the intended cell remains visible. A natural-pair screen separately searches policy-generated continuations for robot-divergent/object-matched and object-divergent/robot-matched futures; only four states meet its criteria, below the planned minimum of ten.

For Cosmos 3, a separately frozen factor study selected 12 population states round-robin by native object-position separation. Ten completed across all six tasks. Starting from the recipient video, an object-priority disjoint mask copies donor object pixels, donor robot pixels outside the object mask, both, or neither. This 2×2 design tests isolated pixel factors; it does not guarantee an in-distribution video or isolate a latent semantic variable.

3.4 K/V interventions

In Cosmos Policy, action queries can attend to future keys. We continuously gate future keys at prespecified block 27, with exact all-key recomputation, equal-count random-key and current-key controls, and a block-0 control. This removal changes attention normalization, so it tests whether a late future-to-action edge is active, not the information carried by a token-count-preserving route.

Cosmos 3 permits the stronger replacement test. We record future-video keys and values from a self-future run, then substitute them for donor-run future-token K/V at the frozen direct interface. Token count, token order, current-video K/V, text K/V, action K/V, observation, instruction, and noise remain fixed. Layer and pathway choices were frozen on excluded calibration tasks before the 22-state population intervention. We report *K/V replacement loss*: the reduction in donor projection after substituting self-future K/V. This is an operational intervention contrast, not a population natural indirect effect.

3.5 Statistical analysis

The independent unit is a restored simulator state. We first average reciprocal directions, donors, noise draws, and process repeats within state, then form 95% percentile intervals from 10,000 bootstrap resamples of states [4]. Task-balanced sensitivity analyses average states within task before resampling tasks. We prespecified 0.10 projection units—10% of the native recipient-to-donor displacement—as the smallest effect of interest for the Cosmos Policy confirmatory family and the Cosmos 3 population and factor families [9]. Prespecified primary sign-test p-values use Holm correction.

4 Results

4.1 Do coherent futures steer behavior?

Coherent donor futures move both the recipient's action and its one-chunk simulator endpoint toward the donor. The effect is strongest at the first queried state.

Cosmos 3. Across 22 completed states and all six tasks, predicted-future action steering is 0.992 (95% state-bootstrap CI $[0.972, 1.010]$), positive in 22/22 states. Relative to the natural-future control, the effect is 0.433 ($[0.332, 0.538]$) larger. Futures encoded from executed donor branches produce nearly the same result: action steering is 0.983 ($[0.890, 1.064]$), and executing the recomputed action yields endpoint steering of 1.003 ($[0.950, 1.050]$). Both are positive in all 22 states. Executed-future action steering also exceeds the geometry-matched Gaussian control by 0.941 ($[0.837, 1.032]$) and the natural control by 0.391 ($[0.246, 0.533]$). Task-balanced analyses preserve these conclusions. A separate four-task engineering study identifies the correct donor in 12/12 cases for both future sources.

Cosmos Policy. At the first query, donor-minus-self action steering is 0.499 ($[0.336, 0.679]$), positive in all ten tasks. The executed endpoint shifts by 0.552 ($[0.387, 0.728]$), also positive in all ten. Action steering remains positive relative to Gaussian (0.497), natural (0.103), and shuffled-future (0.255) controls; the corresponding endpoint contrasts are 0.549, 0.110, and 0.277. Secondary control intervals were not available. The later sampled states show much smaller effects: after three chunks, action and endpoint steering are 0.031 and 0.004; after six, they are 0.027 and 0.025 (Figure 2). Because each time point uses a different state, this pattern is an association with state and timing, not a causal estimate of temporal decay.

No screened branch pair in either policy retained a stable success/failure label across continuation seeds. We therefore interpret these results as donor-directed changes in action and one-chunk endpoint, not changes in closed-loop task completion.

4.2 What future content causes steering?

The content that drives steering differs across checkpoints. Cosmos Policy responds mainly to camera-visible robot motion at early states, whereas neither isolated pixel factor recreates Cosmos 3's whole-future effect.

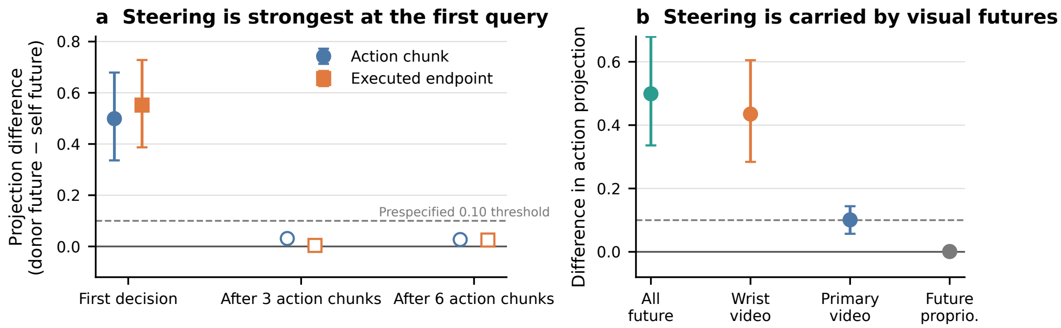

Figure 2: Cosmos Policy results. (a) Donor-minus-self action and endpoint steering is large at first-query states and small at later registered states. (b) At first-query states, visual futures carry the action effect, whereas future proprioception does not. Points are state means; bars are 95% state-bootstrap intervals where reported. The dashed line is the prespecified 0.10 threshold.

Cosmos Policy. Wrist-future action steering is 0.435 ($[0.284, 0.605]$), compared with 0.101 ($[0.057, 0.144]$) for the primary camera and 0.001 ($[-0.004, 0.008]$) for future proprioception. In the 20-state 2×2 study, decoded futures identify the intended cell with 92.5% primary-camera and 98.75% wrist-camera accuracy. Despite this, the object-state main effect is 0.00015 ($[-0.00012, 0.00049]$) for action and 0.00018 ($[-0.00040, 0.00102]$) for the executed goal-state endpoint. The available analysis does not report the robot-state main effect. Four exploratory natural pairs provide a complementary comparison: robot-motion futures yield action steering of 0.291 ($[0.060, 0.521]$) and robot-endpoint steering of 0.315 ($[0.059, 0.571]$), whereas the object-only effects are 0.021 for both action and goal endpoint, with intervals crossing zero. At these sampled early states, the pattern is most consistent with an inverse-dynamics-like role for camera-visible robot motion. It does not show that Cosmos Policy generally ignores object consequences.

Cosmos 3. Across ten factor-study states and all six tasks, the isolated robot-pixel effect is 0.0022 ($[-0.0016, 0.0060]$) for action and 0.0102 ($[-0.0067, 0.0275]$) for the full executed endpoint. The isolated object-pixel effect is 0.0061 ($[0.0009, 0.0119]$) for action and -0.0001 ($[-0.0254, 0.0239]$) for the endpoint. Every interval lies inside the prespecified $[-0.10, 0.10]$ region, and task-balanced estimates agree. Neither factor alone recreates a meaningful fraction of coherent steering (Figure 3c). The effect may depend on information distributed across the coherent future or on nonlinear interactions among content, masks, and tokenization. These experiments do not establish that the two factors are jointly necessary.

4.3 Which internal pathway carries the effect?

Future-token attention provides an active route from generated futures to action. Population-level K/V replacement in Cosmos 3 gives the stronger of the two pathway tests.

Cosmos Policy. Block-27 future-key removal produces a monotonic action-disruption curve: normalized L_2 change is 0.022, 0.035, 0.054, and 0.068 at gates 0.25, 0.50, 0.75, and 1.00. All-key recomputation is bitwise exact. Full removal is 0.016 more disruptive than equal-count random-key removal and 0.048 more disruptive than block-0 future-key removal; equal-count current-key removal is 0.024 larger. Executed endpoints support the same bounded conclusion, although state counts and intervals are not available for this analysis. The late future-to-action edge is active, but it is neither the only nor the largest route through the block.

Cosmos 3. The interface was selected on excluded calibration tasks and frozen before population testing. Replacing donor-run future-token K/V with self-future K/V reduces predicted-future action projection by 0.877 ($[0.842, 0.910]$), with a positive reduction in 22/22 states. For executed futures, the action reduction is 0.814 ($[0.723, 0.900]$) and the executed-endpoint reduction is 0.851 ($[0.766, 0.922]$), again positive in every state. Task-balanced point estimates are 0.877, 0.812, and 0.853; all prespecified primary sign tests remain significant after Holm correction. Because this intervention keeps token count fixed and does not mask attention positions, it shows that the future-

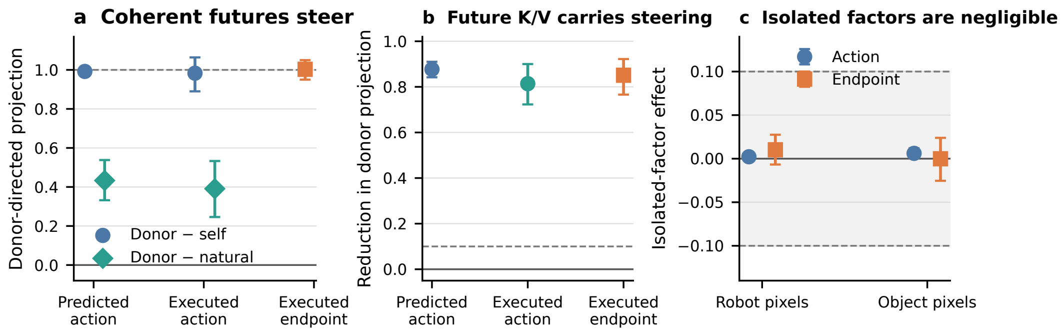

Figure 3: Cosmos 3 population results. (a) Coherent predicted and executed futures steer actions, outperform natural controls, and move one-chunk simulator endpoints toward the donor across 22 states and six tasks. (b) Replacing donor-run future-token K/V with self-future K/V removes most donor-directed action and endpoint projection. (c) In ten factor-study states, isolated robot- and object-pixel effects lie inside the prespecified $[-0.10, 0.10]$ region. Bars are 95% state-bootstrap intervals.

token K/V interface carries donor-directed information. It does not establish a unique or complete circuit.

5 Discussion and Limitations

Across both WAM generations, transplanting a coherent future predicts which alternative action the model will approach. The associated simulator endpoint moves in the same direction. This goes beyond showing sensitivity to generic corruption, but the two checkpoints do not appear to rely on the same content.

Cosmos Policy's early-state pattern resembles an inverse-dynamics interface conditioned on visible robot motion. That account does not transfer cleanly to Cosmos 3: strong whole-future steering coexists with negligible robot-only and object-only pixel effects. Usable content may instead depend on architecture, surrounding context, or distributed feature interactions. The pathway experiments locate where some of this influence travels. Cosmos Policy identifies an active late attention edge, while Cosmos 3 shows across 22 states that future-token K/V carries most of the donor-directed shift.

This evidence is narrower than planning. We do not test candidate generation, outcome comparison, or value-based selection, and we do not establish an effect on binary task success. Here, "future" and "imagined" describe generated representations, not deliberation. Our claim is that these direct policies condition action on coherent futures, not that WAMs generally plan over task consequences.

Limitations. All interventions are in simulation and cover two checkpoints, six RoboLab tasks, and LIBERO-Long. Cosmos Policy's strongest result uses one first-query state per task, while later estimates use different states and cannot isolate elapsed time. Its natural content analysis contains only four states, and rendered hybrids may be outside the policy's native future distribution. Cosmos 3 completed 22 of 24 frozen population states and 10 of 12 factor states; the two population omissions resulted from short source videos and were not replaced. Pixel masks may miss distributed features or induce nonlinear changes, so negligible isolated effects do not establish joint necessity. Finally, key removal changes attention normalization, and K/V replacement can interact with unpatched routes. These interventions identify active pathways, not a full circuit.

Broader impact. Mechanistic tests can help distinguish plausible-looking predictions from representations that actually control behavior. The same interfaces could also redirect robot actions or create false confidence if validated on narrow state distributions. Use beyond simulation would require broader coverage, physical safety constraints, and monitoring outside the intervention metric.

6 Conclusion

In two WAM checkpoints, coherent generated futures steer actions and one-chunk simulator endpoints toward a specific donor continuation. Cosmos 3 shows this effect across 22 sampled states; future-token K/V replacement strongly attenuates it, while isolated robot or object pixels do not recreate it. Cosmos Policy shows a related, more state-dependent effect dominated by visual robot-motion information. The models use their futures to condition action, but whether they compare predicted outcomes to plan remains an open question.

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A Reproducibility Details

The Cosmos Policy experiments use public commit `18a2acca`, checkpoint revision `cb689ec0`, LIBERO, PyTorch 2.7 with CUDA 12.8, and two NVIDIA H100 80GB GPUs. The frozen design contains ten tasks, three state indices, eight native policy runs per state, one reciprocal donor pair per state, and three future-noise draws per direction. Confidence intervals use 10,000 bootstrap resamples of simulator states.

The Cosmos 3 experiments pin Cosmos commit `9aa98e5a`, Cosmos Framework commit `d4599e2e`, RoboLab commit `9db0aaf0`, and checkpoint revision `2b9f9517`. The frozen manifest has SHA-256 `46b6fe3d6abb07712d0926361587b853b2e8313a10c7faeeccb0150d030e986b`. Endpoint evaluation reconstructs the environment, restores the saved state, and replays the observation prefix. Separate controls test token layout, sampler wrapping, cache record and replay, self replacement, and the action-coordinate boundary. Complete timing and aggregate compute accounting remain to be added before submission.

B Additional Results and Boundaries

Across all 30 Cosmos Policy states, mean donor-minus-self steering is 0.186 ($[0.095, 0.290]$) for actions and 0.194 ($[0.091, 0.305]$) for endpoints. Because the three state indices may be nested within task trajectories, the first-query analysis is the cleaner confirmatory result.

The six-state RoboCasa replication is exploratory and is not pooled with LIBERO [13]. Its donor-minus-self endpoint effect is 0.0287 ($[0.0101, 0.0492]$), positive in 6/6 states; action steering is smaller and does not reliably exceed the Gaussian control. In the Cosmos 3 population, the two missing frozen states ended after 21 and 29 actions, before producing the 33-frame source video. Neither was replaced. The prespecified minimum of 20 states across five tasks and the factor-study minimum of ten states across five tasks were both met across six tasks.

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